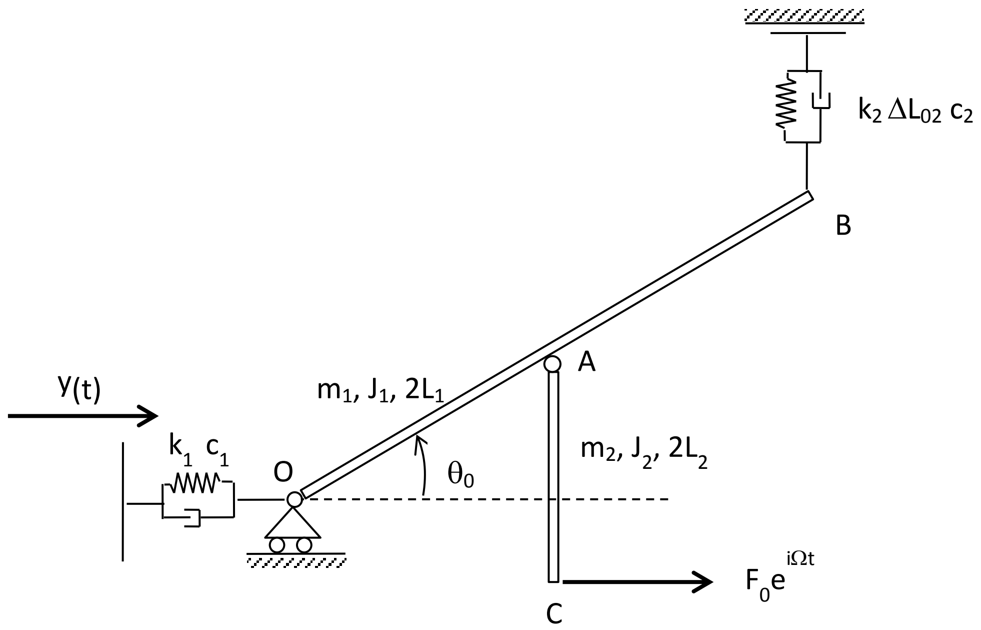
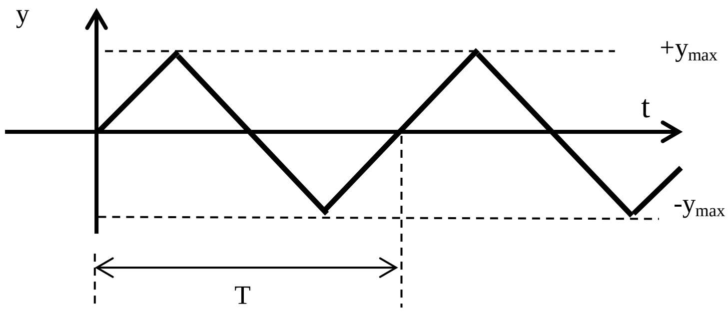
Mechanical Systems Dynamic

February 22nd, 2016



$m_1 = 10$ kg	$k_1 = 500$ N/m
$m_2 = 3$ kg	$k_2 = 500$ N/m
$J_1 = 0.8$ kgm^2	$\Delta L_{02} = 0.1275$ m
$J_2 = 0.2$ kgm^2	$c_1 = 0.2$ Ns/m
$L_1 = 0.5$ m	$c_2 = 0.2$ Ns/m
$L_2 = 0.2$ m	$\theta_0 = \pi/6$ rad
$OA = L_1$	$T = 2.0$ s
$AB = L_1$	$y_{max} = 0.1$ m



The mechanical system above lays in the vertical plane and is shown in its equilibrium position.
The students are requested to:

1. Write the surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet and leave blank the field named *Nr*. The sheet with uncompleted personal data may be not corrected. For environmental reasons the students are also kindly requested to write all of the four pages of each sheet before using another one.
2. Write the linearized equations of motion around the given static configuration;
3. Compute the natural frequencies and modes of vibrations;
4. Compute the Frequency Response Function (in the range 0 – 3 Hz, step = 0.01 Hz) applying the force $F_0 e^{i\Omega t}$. Output: the rotation of the OB bar and the horizontal component of the displacement of point A. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 3 Hz, step = 0.01 Hz) applying an harmonic displacement $y(t) = y_0 e^{i\Omega t}$. Output: the absolute rotation of the AC bar and the force transmitted by the spring and damper system (k_2 c_2). Save the graphical results as SNxxx3.fig and SNxxx4.fig
6. Compute the time history of the horizontal displacement of point C applying the periodical displacement y(t) shown above. Save the results as SNxxx5.
7. Compute the time history of the constraint force of the cart O applying the periodical displacement y(t) shown above. Save the results as SNxxx6.

<u>Note</u>: Please collect the matlab script file relative to the numerical solution of the exercise and save it with the name SNxxx.m, where S states for the first letter of the surname, N for the first letter of the name and xxx are the last three digits of the identification number (e.g. for Bruni Stefano 123456: BS456.m). <u>It is mandatory</u> that all of the files (.fig e .m) have to be compressed (please select either the **.zip** or the **.rar** format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded to the beep portal of Politecnico di Milan. Each student is requested to:

1. Connect to the site https://beep.metid.polimi.it;
2. Enter the portal using its own identification code and password;
3. Select the section *Homework* of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test Feb. 22 2016*
5. Click on the button *Add* and select *Single document*;
6. Browse the compressed file SNxxx.zip using the button *Browse… (or Sfoglia…)*
7. <u>Fill the *Title* field writing its own code SNxxx</u> (e.g. for Bruni Stefano 123456: BS456);

Confirm the procedure clicking on the button *Publish*.